



Quantifying the permissible technical groundwater heat pump potential and its uncertainty through aquifer clustering and spatial packing

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Abstract

Quantifying the permissible technical groundwater heat pump (GWHP) use potential is important for informing policy decisions. This study introduces a transferable and efficient methodology to quantify the permissible technical GWHP use potential and its uncertainty using existing hydrogeological data. The methodology is applied to the deep aquifer in the Baar-Zug-Steinhausen area (Canton of Zug, Switzerland). To address the computational demands of strategies based on 3D numerical groundwater flow and heat transport simulations, the aquifer is classified into distinct hydrogeological “clusters”. This clustering is based on key hydrogeological parameters such as aquifer thickness, permeability, and hydraulic gradient. For each cluster, groundwater flow and heat transport are simulated for a single GWHP doublet in 3D box models with homogeneous properties and a constant hydraulic gradient. Small, medium, and large demands, including both balanced and unbalanced energy load profiles are considered. Thermal and hydraulic influence zones are delineated using 0.1 K isotherms and Darcy flow deviation from natural conditions. A spatial packing algorithm is then applied to place the influence zones within their cluster such that they are aligned with hydraulic gradient direction and treating them as non-overlapping hard boundaries. The permissible technical GWHP use potential is defined by the total number of GWHPs that can be accommodated within each cluster without interference. Uncertainty on the permissible technical potential is quantified through additional simulations. This approach also underscores the importance of quantifying and communicating the range of possible permissible technical heat use potentials to stakeholders, guiding future development towards optimized and environmentally sound groundwater heat use.

Keywords Groundwater heat pump · Technical potential · Heat · Energy · Uncertainty

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Quantifizierung des rechtlich zulässigen technischen Grundwasserwärmenutzungspotenzials und seiner Unsicherheiten mittels Grundwasserleiterklassifizierung und Packalgorithmus

Zusammenfassung

Die Quantifizierung des technischen Nutzungspotenzials von Grundwasser-Wärmepumpen (GWP) ist entscheidend für fundierte politische Entscheidungen. Diese Studie stellt eine übertragbare und effiziente Methodik vor, mit der das technische Potenzial und dessen Unsicherheit auf Basis vorhandener hydrogeologischer Daten quantifiziert werden können. Die Methodik wird auf den tiefen Grundwasserleiter im Gebiet Baar–Zug–Steinhausen (Kanton Zug, Schweiz) angewendet. Um den hohen Rechenaufwand vollständiger 3D-Modelle zu umgehen, wird der Aquifer anhand von Hauptparametern (Mächtigkeit, Permeabilität, hydraulisches Gefälle) in hydrogeologische Cluster eingeteilt. Für jeden Cluster wird die Grundwasserströmung und der Wärmetransport in homogenen 3D-Modellen mit konstantem Gefälle und einem einzelnen GWP-Dublett simuliert. Es werden verschiedene Leistungsanforderungen (klein, mittel, groß) und Energieprofile (ausgeglichen/unausgeglichen) berücksichtigt. Einflusszonen werden über 0,1-K-Isothermen (thermisch) und Abweichungen der Darcy-Strömung (hydraulisch) definiert. Ein Packalgorithmus platziert die Einflusszonen ohne Überlappung im jeweiligen Cluster entlang der Strömungsrichtung. Das technische Nutzungspotenzial ergibt sich aus der maximalen Anzahl an GWP-Dubletts pro Cluster ohne gegenseitige Beeinflussung. Die Unsicherheit wird durch zusätzliche Simulationen quantifiziert. Die Ergebnisse unterstreichen die Bedeutung einer transparenten und nachvollziehbaren Berechnung verschiedener technischer Wärmenutzungspotenziale sowie ihrer Unsicherheiten – sowohl zur Beratung von Entscheidungsträgern als auch zur Unterstützung einer nachhaltigen und optimierten Nutzung der Grundwasserwärme.

Schlüsselwörter Grundwasser-Wärmepumpen · Technisches Potenzial · Wärmenutzung · Energie · Unsicherheit

Introduction

Quantifying the heat potential of groundwater for GWHP applications is a complex endeavor that requires a thorough understanding of hydrogeological conditions and advanced modeling techniques. The main assumption behind GWHP potential is the need for effective management of thermal interference, which is paramount for the long-term sustainability of GWHP systems. This includes preventing both intra-system thermal short-circuiting and inter-system thermal cascade effects through optimal sizing and placement of extraction-injection well doublets. Therefore, understanding of the thermal and hydraulic impacts of GWHP doublets on groundwater is key.

Aquifer Thermal Energy Storage (ATES) systems play a crucial role in this context, enabling seasonal thermal balancing. By optimizing heat recovery and mitigating seasonal demand mismatches, ATES significantly enhances overall GWHP efficiency and sustainability. Quantifying uncertainty in GWHP potential assessments is important for guiding informed decisions. This process enables stakeholders to evaluate different exploration and development scenarios, considering the probabilities of various outcomes and their associated economic consequences.

Due to the high computational demands of city-scale well placement optimization with 3D groundwater flow and transport models (Stemmle et al. 2024), few studies have sought to simplify the problem through the use of faster, reduced-complexity box-model simulations.

The Thermal Aquifer Potential (TAP) method proposed by Böttcher et al. (2019) assesses the spatial technical potential for the thermal use of groundwater based on empirical relations for maximum sustainable flow rates from well doublets subject to constraints on drawdown, injection water rise and hydraulic breakthrough. These empirical relations were derived from numerical, steady-state, 2D box-model simulations. The TAP method requires the installed wells to be aligned with the groundwater flow direction. Epting et al. (2020) applied the TAP method to quarters of the city of Basel, where the parameters of the 2D box-model simulations were inferred from a 3D groundwater flow and heat-transport model at the city scale.

Halilovic et al. (2024) formulated an optimization framework integrating the TAP methodology within a mixed-integer linear programming (MILP) model, aiming to maximize extractable thermal potential subject to both regulatory and engineering constraints.

Halilovic et al. (2023) modified the 2D linear advective heat transport model (LAHM) to account for (time-varying) energy load profiles and to optimize the positioning of GWHP doublets under technical and regulatory constraints. The optimization was performed with homogeneous hydrogeological parameters including hydraulic gradient.

To assess the technical potential of low-temperature aquifer thermal energy storage (LT-ATES), Stemmle et al. (2024) divided an aquifer at the city scale into four areas with similar hydrogeological settings. They estimated the thermal affected zones (TAZ) of different ATES configurations with numerical 3D box-model simulations and ran

one city-scale simulation of one type of ATEs system with a regional groundwater flow and heat transport model. The ATEs systems were heuristically placed along the groundwater flow direction and separated by the length of their TAZ.

Pelzer and Schulte (2024) developed a two-stage convolutional neural network, initially trained on individual heat plume simulations and subsequently refined using scenarios involving multiple interacting plumes. The steady-state simulations used for training were conducted with a two-dimensional box-model of groundwater flow and heat transport.

These findings underline the potential of proxy simulations based on computationally efficient 3D box-models of GWHP systems. To date, however, no automated mapping of such simulations to city-scale or regional aquifers has been developed. Moreover, the literature emphasizes the uncertainties inherent in the proxy nature of box-model simulations, underscoring the need for uncertainty quantification in heat potential estimations. The remaining challenge lies in developing methods that can efficiently and automatically quantify heat potential and its associated uncertainty from 3D box-model simulations without imposing an excessive computational burden.

In this context, we developed a new methodology to quantify the potential for GWHP doublets for three different but realistic energy load profiles, balanced and unbalanced, using available hydrogeological data.

The remainder of this paper is structured as follows. The Methodology section presents the study site, defines the GWHP heat-use potential, describes the configuration of doublet systems and load profiles, outlines the modeling strategy, and details the approach to uncertainty quantification. The Results section reports the thermal plume dynamics, the simulation of the GWHP doublets, and the resulting heat-use potential. The Discussion section examines the strength and potential improvement of the methodology, and the Conclusion section summarizes the main outcomes.

Methodology

Study site

In the Baar-Zug-Steinhausen area (canton of Zug, Switzerland), beneath a shallow groundwater layer lies a thick deep aquifer, consisting of Quaternary loose sediments at depths of 80 m to 230 m, extending over an area of approx. 10 km² and covered by low-permeability lake deposits. The artesian, confined deep groundwater has low groundwater flow velocities and a relatively constant temperature, which are favorable conditions for seasonal heat storage. Near-surface groundwater resources in the canton of Zug are already be-

ing used intensively, and due to water protection regulations, the potential for near-surface groundwater heat pump use is locally nearly exhausted. To estimate the potential for GWHP development in the Baar-Zug-Steinhausen area, it was necessary to quantify the permissible technical GWHP use potential of the deep aquifer and its associated uncertainty. As no regional groundwater model was available, the Office for Environment of the Canton of Zug provided maps of the hydraulic conductivity, aquifer thickness, depth of the aquifer top, and groundwater heads for the defined investigation perimeter (Fig. 1). These maps were created based on the interpolation of results from previous geotechnical and hydrogeological investigations.

Defining GWHP heat use potential

In this study, the heat use potential of groundwater is defined as the maximum number of GWHP systems that can be installed within a designated area without causing significant mutual thermal interference, such as thermal short circuits and cascade effects, and without contravening the prevailing Swiss regulations concerning groundwater temperature changes. In Switzerland, the groundwater temperature must not deviate by more than 3 °C at a distance of 100 m from the reinjection point from the natural, undisturbed groundwater temperature.

Therefore, the proposed heat use potential represents the maximum permissible technical groundwater heat use potential achievable under ideal conditions (no thermal interference), abstracting from existing infrastructure or other site-specific usage restrictions. It is defined in terms of power (in MW) or specific power (W/m²).

Doublet systems and monthly energy load profiles

This potential is primarily governed by the intrinsic hydrogeological properties of the aquifer as well as by the design and operational parameters of the GWHP system itself. The objective is to estimate the GWHP potential for three types of unbalanced and balanced monthly energy load profiles representing small, medium, and large energy demands as defined in Jörin et al. (2022). The unbalanced energy load profiles are based on real data:

- Energy load profile 1: Multi-family house with a heating capacity of 60 kW for space heating and domestic hot water production. 11% of the extracted groundwater heat is reinjected for cooling purposes. The maximum pumping rate is about 68 m³/d.
- Energy load profile 2: Commercial and administrative building with a heating capacity of 250 kW for space heating and domestic hot water production. 44% of the

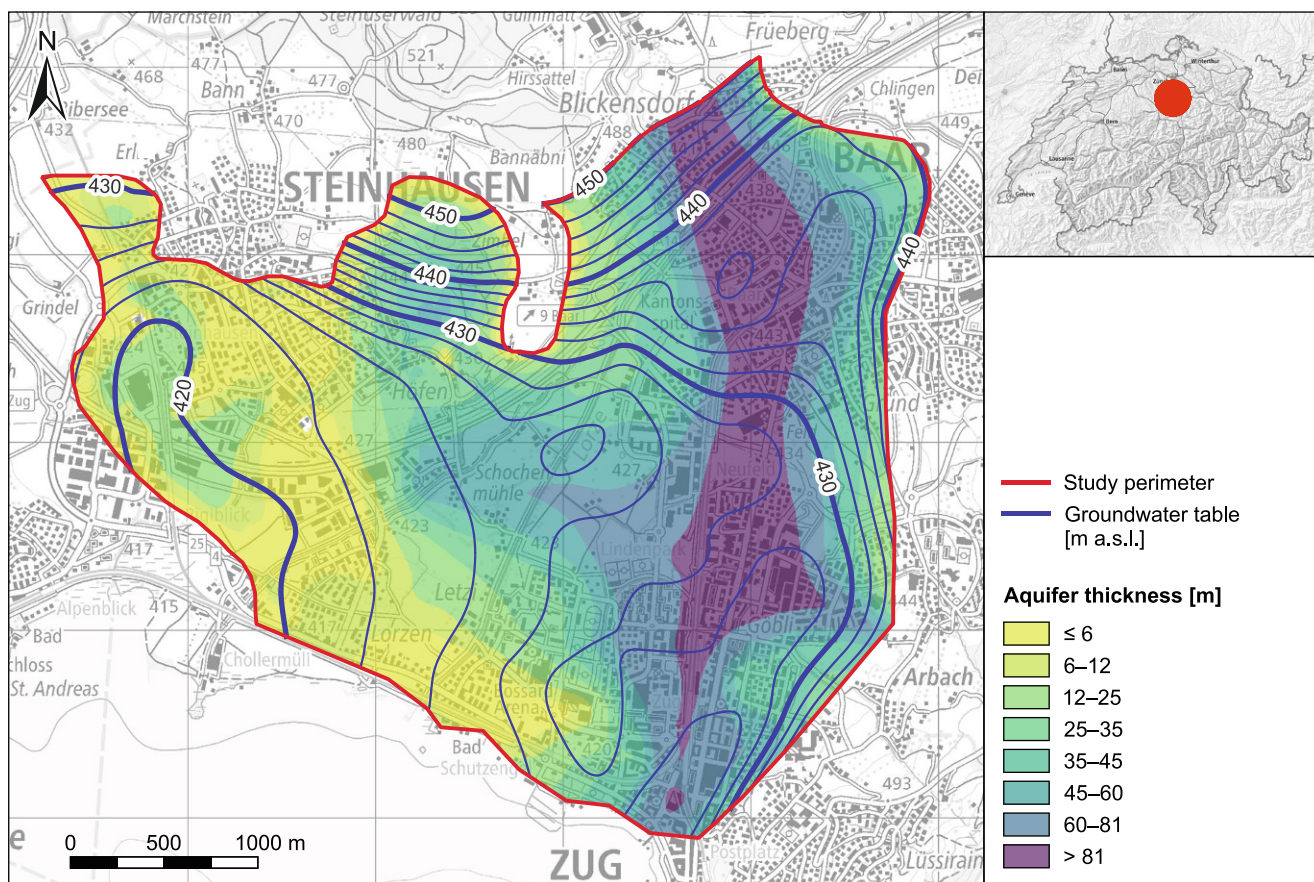


Fig. 1 Study perimeter with groundwater head isolines and aquifer thickness

Abb. 1 Untersuchungsperimeter mit Mächtigkeit und Grundwasserisohypsen des Grundwasservorkommens

extracted groundwater heat is reinjected for cooling purposes. The maximum pumping rate is about 242 m³/d.

- **Energy load profile 3:** Office building with a heating capacity of 375 kW for space heating and domestic hot water production. 44% of the extracted groundwater heat is reinjected for cooling purposes. The maximum pumping rate is about 364 m³/d.

The balanced energy load profile is derived from the unbalanced profile by introducing an energy source during the summer months, which could, for example, be supplied by solar thermal energy, a local energy network, or district heating.

Figure 2 shows the three energy load profiles. For simplification, doublet systems are simulated, which consist of a pair of wells for groundwater extraction and injection set 100 m apart and lying on the same groundwater flow line. The operational mode of these doublets, whether bidirectional or recirculative, is strategically chosen based on the prevailing hydraulic gradient. Bidirectional systems, in which the extraction and reinjection wells switch roles seasonally, are more effective for heat recovery in areas with low gradients (typically less than 1%), whereas recircula-

tive doublets, in which the extraction well is located downstream of the injection well, are employed in regions with higher gradients. The temperature difference between injection and extraction is set to 8 K for both cooling ($\Delta T_{\text{cooling}}$) and heating ($\Delta T_{\text{heating}}$). For months when both cooling and heating are required, the temperature difference between injection and extraction ΔT and the monthly pumping rate (Q) are calculated according to energy conservation as follows:

$$\Delta T = \frac{Q_{\text{cooling}} |\Delta T_{\text{cooling}}| - Q_{\text{heating}} |\Delta T_{\text{heating}}|}{\max(Q_{\text{cooling}}, Q_{\text{heating}})} \quad (1)$$

$$Q = \max(Q_{\text{cooling}}, Q_{\text{heating}}) \quad (2)$$

Q_{cooling} and Q_{heating} denote the pumping rates for cooling and heating, respectively.

Strategy

To accurately assess GWHP potential and account for thermal interference, numerical simulations that model the coupled groundwater flow and heat transport processes over extended operational periods (often spanning 20 years) are

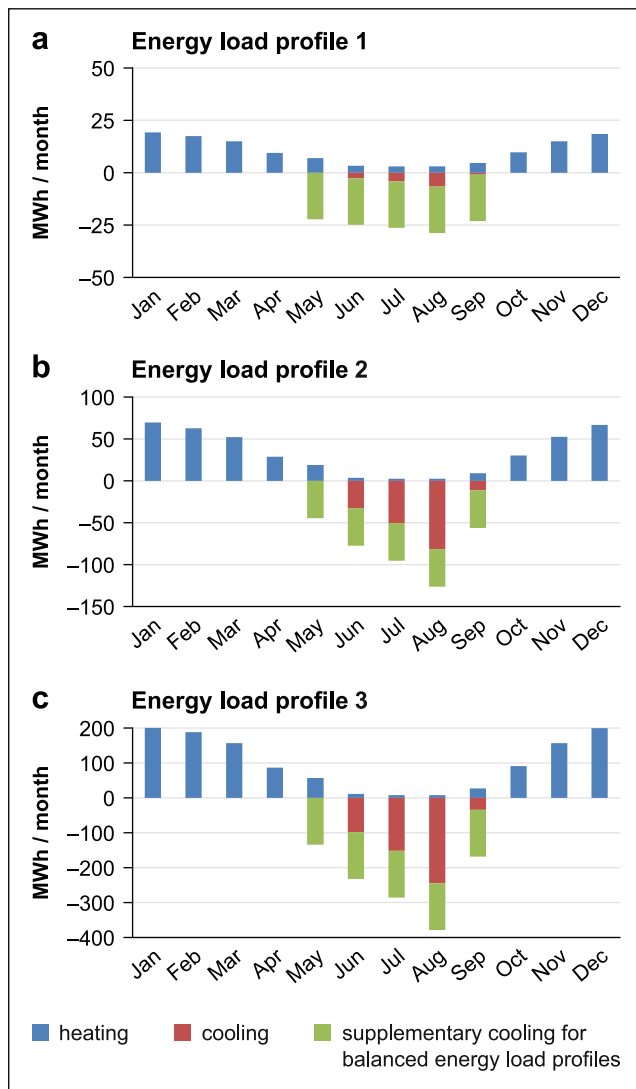


Fig. 2 **a** Energy load profile 1 (multi-family house, 60 kW heating capacity). **b** Energy load profile 2 (commercial and administrative building, 250 kW heating capacity). **c** Energy load profile 3 (office building, 375 kW heating capacity)

Abb. 2 **a** Energielastprofil 1 (Mehrfamilienhaus, 60 kW Spitzenleistung), **b** Energielastprofil 2 (Gewerbe- und Verwaltungsbau, 250 kW Spitzenleistung) und **c** Energielastprofil 3 (Bürobau, 375 kW Spitzenleistung)

indispensable. The iterative modeling of numerous GWHP doublets—aimed at minimizing thermal interference, ensuring regulatory compliance, and maximizing the number of GWHP doublets—presents a significant computational challenge. To mitigate this, a simplified approach is proposed, which is also suitable in cases where no groundwater model is available:

1. Spatially classify the aquifer into representative hydrogeological units or “clusters”.

2. For each identified cluster, extract the representative median values for hydraulic conductivity, groundwater thickness, and hydraulic gradient.
3. For each cluster, simulate the GWHP doublet with a 3D numerical groundwater flow and heat transport box-model with homogeneous properties and constant hydraulic gradient corresponding to the representative median values.
4. Check whether the simulated GWHP doublet complies with Swiss legal regulations (i.e., verify that the temperature change 100 m downstream of the reinjection point is less than 3 K). If it does not, reduce the power of the GWHP doublet until the regulation is satisfied.
5. Compute the heat recovery efficiency.
6. Delineate the thermal and hydraulic influence zones using Darcy flow vectors and 0.1 K isotherms (i.e., thermal plume).
7. Estimate the temperature deviation 100 m downstream the reinjection well.
8. Apply a simple packing algorithm to place the influence zones within each cluster.
9. The number of GWHP doublets indicates the heat potential as well as the recovery heat potential

Details are given below.

The aquifer classification is achieved through the k-Means Clustering algorithm (Press et al. 1992) applied to the following hydrogeological parameters: hydraulic conductivity, groundwater thickness, and hydraulic gradient. These parameters were available as a raster dataset; the hydraulic gradient was derived from the groundwater heads. Several possible numbers of clusters were tested. Five clusters were chosen because the clusters are well connected (i.e., they form roughly continuous areas), are large enough, and are representative of distinct hydrogeological settings (i.e., the hydrological properties of the clusters are significantly distinct, see supplementary material). Note that this is a subjective choice. However, the algorithm still produces some small and unfavorable polygons that are unsuitable for further calculations. Therefore, these polygons were assigned to other clusters to obtain a more homogeneous picture (see supplementary material).

For each adjusted cluster, the median value of the individual parameters was assumed as a representative value and used for the thermal numerical simulations (Table 1).

For each cluster and each energy load profile, the GWHP doublet was simulated with both balanced and unbalanced energy load profiles over a 20-year operating period. Groundwater flow and heat transport were modeled in FEFLOW within a box model measuring 2000 m in length and 500 m in width. FEFLOW, which is based on the finite element method, simulates groundwater flow using the saturated Darcy equation and heat transport with

Table 1 Median values per cluster and mean values per subcluster for hydraulic gradient, hydraulic conductivity, and aquifer thickness (saturated thickness). The subclusters are defined in “Section Uncertainty” quantification.

Tab. 1 Medianwerte pro Cluster und Mittelwert pro Subcluster für den hydraulischen Gradient, die hydraulische Durchlässigkeit und die Grundwassermächtigkeit. Die Subclusters sind im Kapitel “Uncertainty quantification” definiert.

Cluster/Subcluster	Gradient [%]	Hydraulic conductivity [m/s]	Aquifer thickness [m]
<i>Cluster 1 (bidirectional)</i>	0.57	$9.4 \cdot 10^{-5}$	54
Subcluster 1.1	0.66	$9.3 \cdot 10^{-5}$	48
Subcluster 1.2	0.63	$9.76 \cdot 10^{-5}$	76
<i>Cluster 2 (bidirectional)</i>	0.47	$3.4 \cdot 10^{-5}$	8
Subcluster 2.1	0.49	$5.78 \cdot 10^{-5}$	23
Subcluster 2.2	0.53	$3.54 \cdot 10^{-5}$	6
<i>Cluster 3 (recirculative)</i>	2.1	$3.6 \cdot 10^{-5}$	34
Subcluster 3.1	1.96	$1.86 \cdot 10^{-5}$	14
Subcluster 3.2	2.24	$5.8 \cdot 10^{-5}$	42
<i>Cluster 4 (recirculative)</i>	3.68	$4.4 \cdot 10^{-5}$	29
Subcluster 4.1	3.5	$1.26 \cdot 10^{-5}$	10
Subcluster 4.2	3.72	$9.84 \cdot 10^{-5}$	35
<i>Cluster 5 (bidirectional)</i>	0.54	$2 \cdot 10^{-4}$	84
Subcluster 5.1	0.8	$1.9 \cdot 10^{-4}$	88
Subcluster 5.2	0.63	$3.95 \cdot 10^{-4}$	25

the advection-dispersion equation. Heat transport includes conduction within the porous medium, advection by flowing groundwater, thermal dispersion, and heat exchange between the solid matrix and the fluid. The box models are inclined along the groundwater gradient and simplified into three geological layers (see also Table 2):

- Confining layer: low permeability, discretized into layers of increasing thickness by a factor of 1.5 in the upward direction, with the bottom layer being 3 m thick (maximum 10 layers).
- Aquifer: medium to high permeability (Table 1), thickness depends on the cluster, discretized into 2 m thick layers
- Bedrock (aquitard): low permeability, discretized into four layers of increasing thickness by a factor of 1.5 in the downward direction, with the top layer being 3 m thick.

Each layer consists of 7904 tetrahedral elements, with element sizes ranging from 0.4 to 40 m. The simulation parameters are given in Table 2 and were derived from literature values in accordance with the Office for Environment of the Canton of Zug.

It is assumed that the groundwater is confined. The doublet wells are placed 100 m apart for all simulations and fully penetrate the aquifer. The hydraulic gradient is imposed using first-type (Dirichlet) hydraulic head boundaries at the inflow and outflow sides of the box model, with the hydraulic head at the outflow side set equal to the surface elevation. All other flow model boundaries are defined as impervious. All thermal boundaries are assigned as zero temperature-gradient conditions.

Since no groundwater temperature data were available and only temperature changes caused by system operation are relevant to the present research question, natural sea-

sonal variations in groundwater temperature were not considered. The simulation therefore focuses solely on the relative temperature change induced by the planned systems, with both the initial and reference temperatures in FEFLOW set to 0 °C.

The heat recovery efficiency metric quantifies how effectively stored thermal energy is retrieved in subsequent operational cycles. It is calculated as the ratio of the energy extracted in the current cycle to the energy stored in the preceding cycle (e.g., Abuasbeh et al. 2021):

$$\eta_i = \frac{E_i^\uparrow}{E_{i-1}^\downarrow} = \frac{\int_{t_i}^{t_{i+1}} Q_i \rho c (T_i^\uparrow - T_{GW}) dt}{\int_{t_{i-1}}^{t_i} Q_{i-1} \rho c (T_{i-1}^\downarrow - T_{GW}) dt} \quad (3)$$

with Q_i the extraction rate in cycle i (m³/s), c the specific heat capacity of water (J/kg/K), ρ the density of water (kg/m³), $T_i^\uparrow$ the temperature (°C) of the extracted groundwater in cycle i , $T_{i-1}^\downarrow$ the temperature (°C) of the injected groundwater in cycle $i-1$, T_{GW} the natural temperature (°C) of the groundwater, t_{i-1} the start time of cycle $i-1$, and t_i the start time of cycle i . Since only the relative temperature change caused by the planned systems is simulated, T_{GW} is set to zero. In heating cycles, energy is only calculated if $T_i^\uparrow > T_{GW}$ and $T_{i-1}^\downarrow > T_{GW}$. In cooling cycles, energy is only calculated if $T_i^\uparrow < T_{GW}$ and $T_{i-1}^\downarrow < T_{GW}$. For each simulation, the efficiency of heat recovery was averaged over the simulation period, excluding the first and last cycles (see graphical explanation in supplementary material).

The *influence zone* of the GWHP doublet is defined by the convex hull of both (i) the 0.1 K isotherm, and (ii) the area where the Darcy velocity is at least five times greater than the background velocity (see supplementary material). The background velocity refers to the Darcy velocity that

Table 2 Hydraulic and thermal parameters of the simplified 3D thermal groundwater flow model**Tab. 2** Hydraulische und thermische Parameter des vereinfachten thermischen 3D-Grundwasserströmungsmodells

Parameter	Value
<i>Hydraulic gradient</i>	Cluster specific
<i>Confining layer</i>	
Hydraulic conductivity	$1 \cdot 10^{-7}$ m/s
Porosity	0.15
Volumetric heat capacity	2.0 MJ/m ³ /K
Thermal conductivity of solid	1.8 W/m/K
Longitudinal dispersivity	20 m
Lateral dispersivity	5 m
Thickness	~35.76 m
<i>Aquifer</i>	
Hydraulic conductivity	Cluster specific
Porosity	0.15
Volumetric heat capacity	2.3 MJ/m ³ /K
Thermal conductivity	1.7 W/m/K
Longitudinal dispersivity	20 m
Lateral dispersivity	5 m
Thickness	Cluster specific
<i>Bedrock (aquitard)</i>	
Hydraulic conductivity	$1 \cdot 10^{-7}$ m/s
Porosity	0.15
Volumetric heat capacity	2.1 MJ/m ³ /K
Thermal conductivity	2.6 W/m/K
Longitudinal dispersivity	20 m
Lateral dispersivity	5 m
Thickness	~40 m

would occur in the absence of the GWHP system. The velocity criterion was determined after testing to prevent hydraulic interference between the GWHP systems and to ensure, for recirculative doublets, that the extraction well and its influence area lies within the influence zone.

The *spatial packing* algorithm employs an iterative procedure to maximize the number of GWHP doublets placed within a defined cluster area. This is done under strict adherence to the following conditions: (i) the influence zones must not overlap—thus avoiding thermal cascading effects; (ii) they must stay within their assigned clusters, although overlap with other cluster areas is permitted; and (iii) they must be aligned with the flow direction. The algorithm begins by placing the largest influence zones within the cluster area based on a predefined raster grid. Only influence zones that satisfy all placement criteria are retained. Subsequently, the algorithm performs two types of updates that are inspired by simulations of point processes (e.g., Illian et al. 2008): move updates, where existing influence zones are randomly repositioned, and birth updates, where new influence zones are introduced into the available free space. Each update is accepted only if the resulting configuration

continues to satisfy all placement constraints. The algorithm was run for several hours, and the iteration yielding the largest number of influence zones was selected.

Uncertainty quantification

In the present study, the uncertainty quantification of the heat use potential is limited to: (1) hydrogeological parameters—namely, hydraulic conductivity, groundwater thickness, and hydraulic gradient (Table 1) and (2) the alignment of the doublet with the groundwater flow direction.

To limit the number of simulations, uncertainty in the hydrogeological parameters is addressed by dividing each cluster into two subclusters using the k-means algorithm (see supplementary material). The mean values of the hydrogeological parameters—hydraulic conductivity, groundwater thickness, and hydraulic gradient—within each subcluster are considered representative of the uncertainty (the mean is preferred over the median as it yields more variable results). Additionally, for each subcluster, one extra simulation is performed with a 6° shift in downstream well orientation relative to the natural groundwater flow direction, corresponding to approximately 17 m for a well spacing of 100 m. This shift reflects the practical reality that wells are often not perfectly aligned with the groundwater flow direction.

Following the same approach described earlier, influence zones are estimated for each subcluster and for each orientation angle, resulting in a total of four influence zones per cluster. For each cluster and energy load profile, the smallest and largest influence zones among the four subcluster simulations and the original cluster simulation are identified. The spatial packing algorithm is then applied using both the smallest and the largest influence zones to estimate the range of the heat use potential.

Results

Thermal plume dynamics

The thermal plume dynamics of bidirectional and recirculative GWHP doublets (Rosen and Koochi-Fayegh 2017) are briefly presented to provide a clearer understanding of the results.

Bidirectional GWHP doublets operate by seasonally reversing flow, creating alternating warm and cold plumes that remain mostly spatially separated, with thermal mixing occurring primarily at the interface between the plumes. These plumes, together with the velocity criterion, define the influence zone. Thermal reciprocity allows previously injected energy to be recovered, leveraging the aquifer thermal memory. Efficiency is particularly high in low-gradient

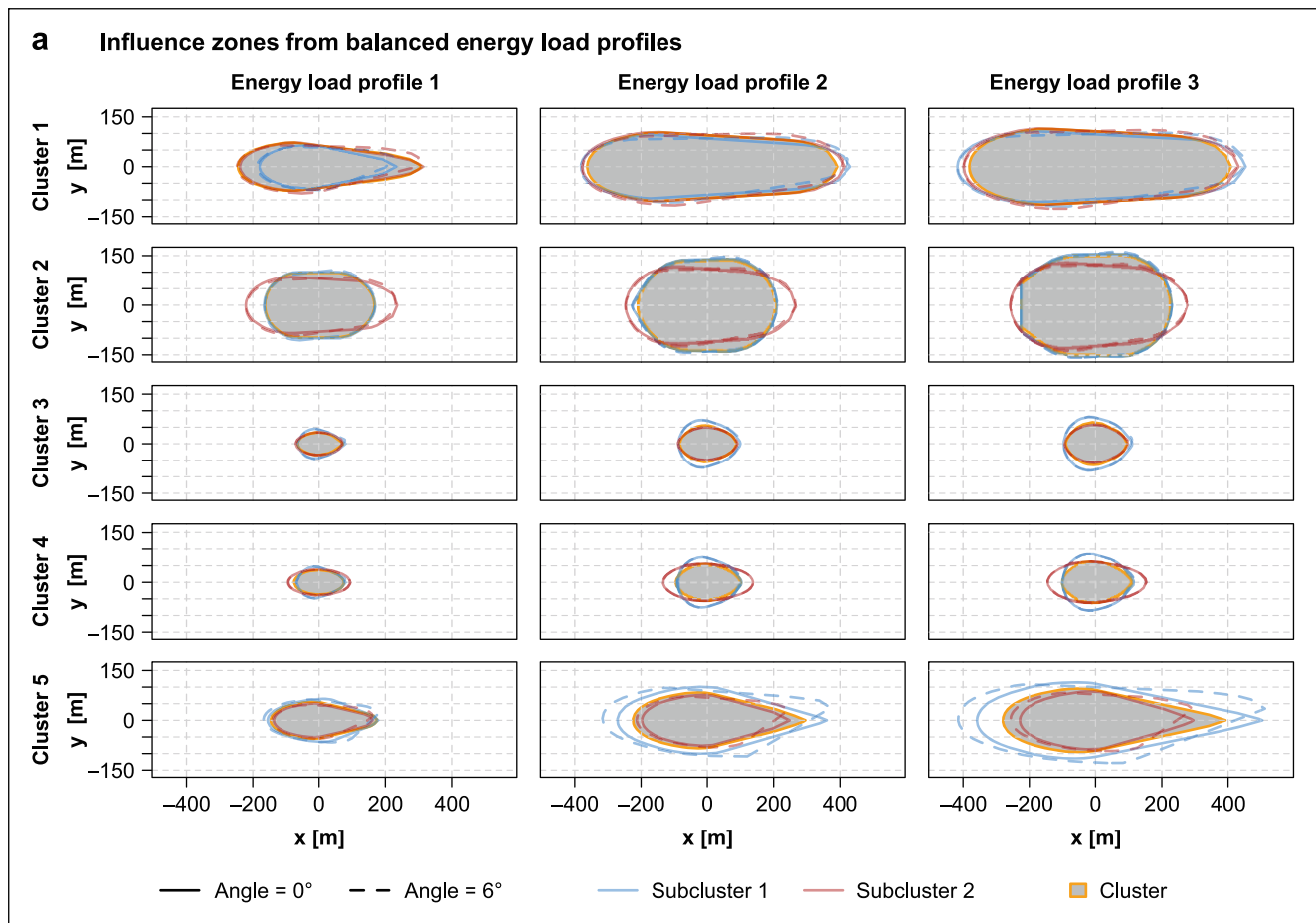


Fig. 3 Influence zones of the clusters and subclusters for the three balanced (a) and unbalanced (b) energy load profiles

Abb. 3 Einflusszonen der Cluster und Subcluster für die drei ausgeglichenen (a) und unausgeglichenen (b) Energielastprofile

aquifers, where plume displacement and convection with regional groundwater flow are minimal, maintaining compact plumes and reducing conductive and dispersive losses (Stemmler et al. 2022).

In recirculative GWHP doublets, injected cold and warm water forms a mixed cold-warm plume advected down-gradient by regional groundwater flow. The single plume, together with the velocity criterion, defines the influence zone. Heat is dissipated through (1) thermal mixing between successive cycles (seasonal thermal neutralization), (2) convection with groundwater flow, and (3) conduction into the aquifer matrix, which absorbs and releases heat from the plume over the heating and cooling cycles. Hydrodynamic dispersion, which increases with velocity (Diersch 2014), enhances convective mixing. Heat recovery is enhanced by an optimal distance between the injection and extraction wells, such that thermal breakthrough is delayed by one cycle.

Simulations of the GWHP doublets

The determined influence zones for the clusters and their subclusters are shown in Fig. 3.

For the balanced energy load profiles, the following can be noted:

- Despite higher groundwater gradients, the recirculative GWHP doublets (cluster 3 and 4) exhibit the smallest influence zones. This should not be interpreted as higher heat recovery (see discussion below). The underlying mechanism is heat dissipation as explained in the Section “Thermal plume dynamics”. Due to a non-negligible groundwater flow velocity, the thermal plumes of bidirectional GWHP doublets are advected downstream, resulting in seasonal thermal neutralization.
- The shift of the well downstream to the natural flow direction (angle = 6°) has an overall minor effect. Recirculative GWHP doublets (Cluster 3 and 4) are minimally affected, as the size of the influence zone is mainly determined by the plume injected into the upstream well

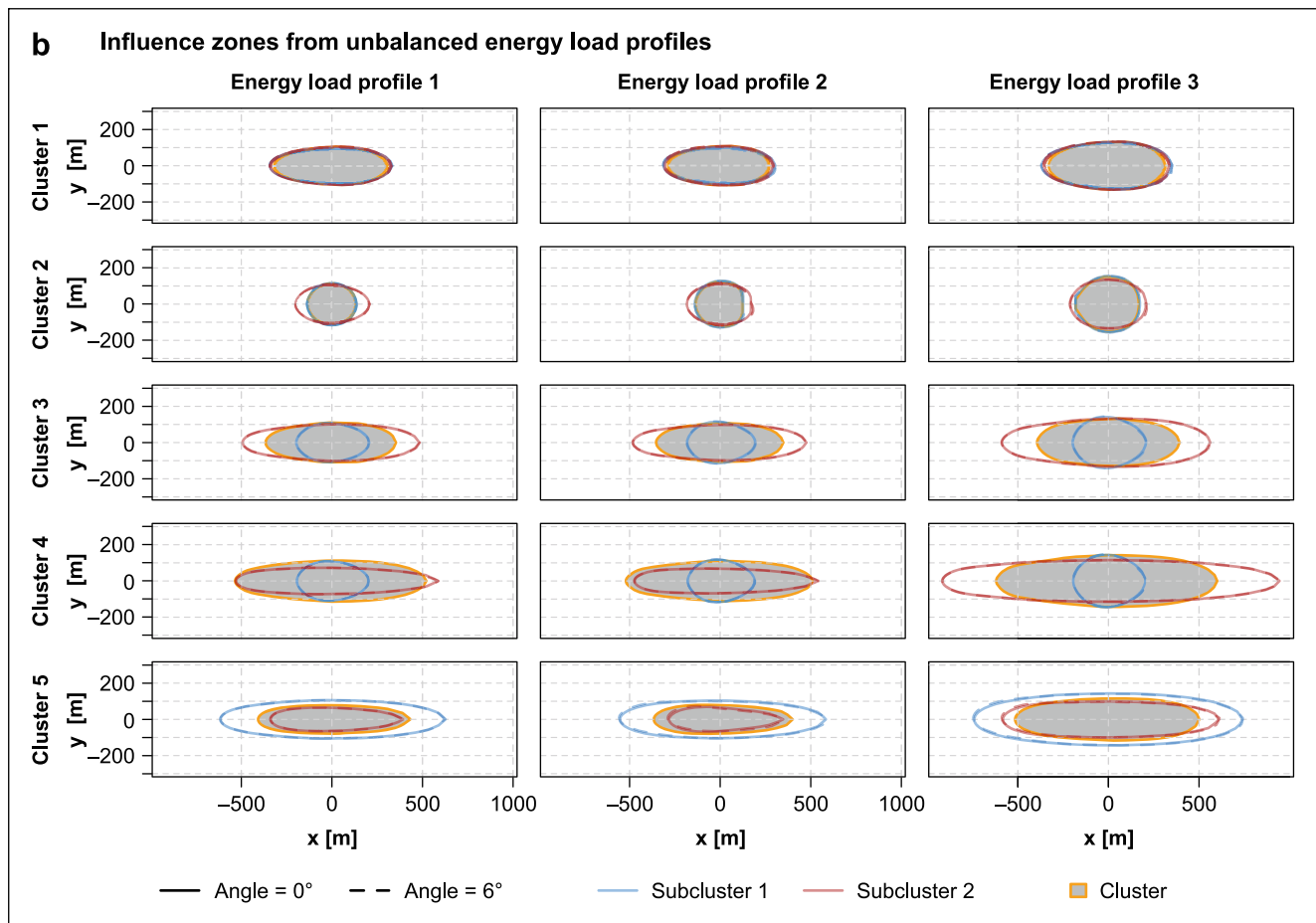


Fig. 3 Influence zones of the clusters and subclusters for the three balanced (a) and unbalanced (b) energy load profiles (Continued)

Abb. 3 Einflusszonen der Cluster und Subcluster für die drei ausgeglichenen (a) und unausgeglichenen (b) Energielastprofile (Fortsetzung)

- and the downstream extraction well exerts only a limited influence on the thermal plume due to the high hydraulic conductivity and gradient. In contrast, the thermal plumes of bidirectional GWHP doublets show cluster-dependent sensitivity: in Clusters 1 and 2, lower groundwater velocities promote greater lateral rather than longitudinal spreading, rendering the plumes less affected by downstream well displacement. In Cluster 5, higher groundwater velocities result in more longitudinal spreading, making the plumes slightly more sensitive to shifts.
- The uncertainty of the influence zones per cluster is described by the size variation of the influence zones between the cluster, its subcluster and the shift of the downstream well. In Cluster 1, the uncertainty decreases with higher energy load profiles, meaning that the heat spread is dominated by the pumping rate (the uncertainty in the hydrogeological parameters is small). Cluster 5 shows increasing uncertainty due to higher velocities and hydrogeological variability. Cluster 2 exhibits slightly decreasing length uncertainty and slightly increasing width

uncertainty with higher loads. Clusters 3 and 4 show no significant change between profiles 2 and 3.

For the unbalanced energy load profiles, the following can be noted:

- The influence zones of the recirculative GWHP doublets (Cluster 3 and 4) become significantly larger compared to the balanced energy load profiles, because the seasonal thermal neutralization is unbalanced, allowing the cold part of the plume to propagate farther downstream. The bidirectional doublets of Cluster 1 and 2 are generally smaller because of the smaller downstream warm plume. The smaller downstream warm plume from the bidirectional doublets of Cluster 5 reduces the seasonal thermal neutralization and therefore strongly enhances the spreading of the cold plume.
- The shift of the well downstream to the natural flow direction (angle=6°) has an insignificant effect on the influence zones, even by Cluster 5. Because of the unbal-

anced energy load profile, the warm downstream plume is smaller and enveloped by the spreading cold plume.

- The largest uncertainties are for Clusters 3, 4, and 5 because of the higher groundwater flow velocities.

For comparison purposes, all the clusters were run for balanced and unbalanced energy load profiles as well as for both operating modes (recirculative and bidirectional). Figure 4 compares the length of the influence zones as well as the heat recovery efficiency for both operating modes.

For unbalanced energy load profiles, recirculative GWHP doublets display slightly larger influence zones than bidirectional doublets (all circles lie just below the 1:1 line in Fig. 4). Under balanced profiles, the influence zones of recirculative doublets are—except in Cluster 2—markedly smaller, as warm and cold thermal plumes are reinjected into the same well and dissipate more effectively through mixing (see Section: “Thermal plume dynamics”). In Cluster 2, this difference is less pronounced for load profiles 2 and 3, since the influence zone size is primarily controlled by the groundwater flow velocity. This results from Cluster 2 having the lowest hydraulic gradient, conductivity, and aquifer thickness.

The heat recovery efficiency of recirculative GWHP doublets is generally low (up to a maximum of 9%) because of the strong thermal dissipation. In the case of unbalanced load profiles, the long-term heat recovery efficiency is governed by the dominant storage mode, which in cooling-dominated systems corresponds to cold storage. Under these conditions, the temperature of the extracted groundwater during the heating season progressively decreases over consecutive cycles, eventually falling below the ambient groundwater temperature. Consequently, almost no warm water can be recovered after a certain number of cycles, indicating that effective seasonal heat storage is no longer achieved. In bidirectional systems, the heat recovery efficiency tends to be higher even if the groundwater flow velocities are not ideal for bidirectional GWHP doublets. The reason for this is the reduced thermal dissipation due to the more efficient use of the aquifer thermal memory.

The legal requirement, which stipulates that the maximum temperature change 100 m downstream of the system must not exceed 3 K, is satisfied in all simulations. In fact, the maximum temperature change at this distance remains below 1.51 K across all cases, indicating that GWHP doublets with capacities greater than 375 kW are permissible under the current legal framework.

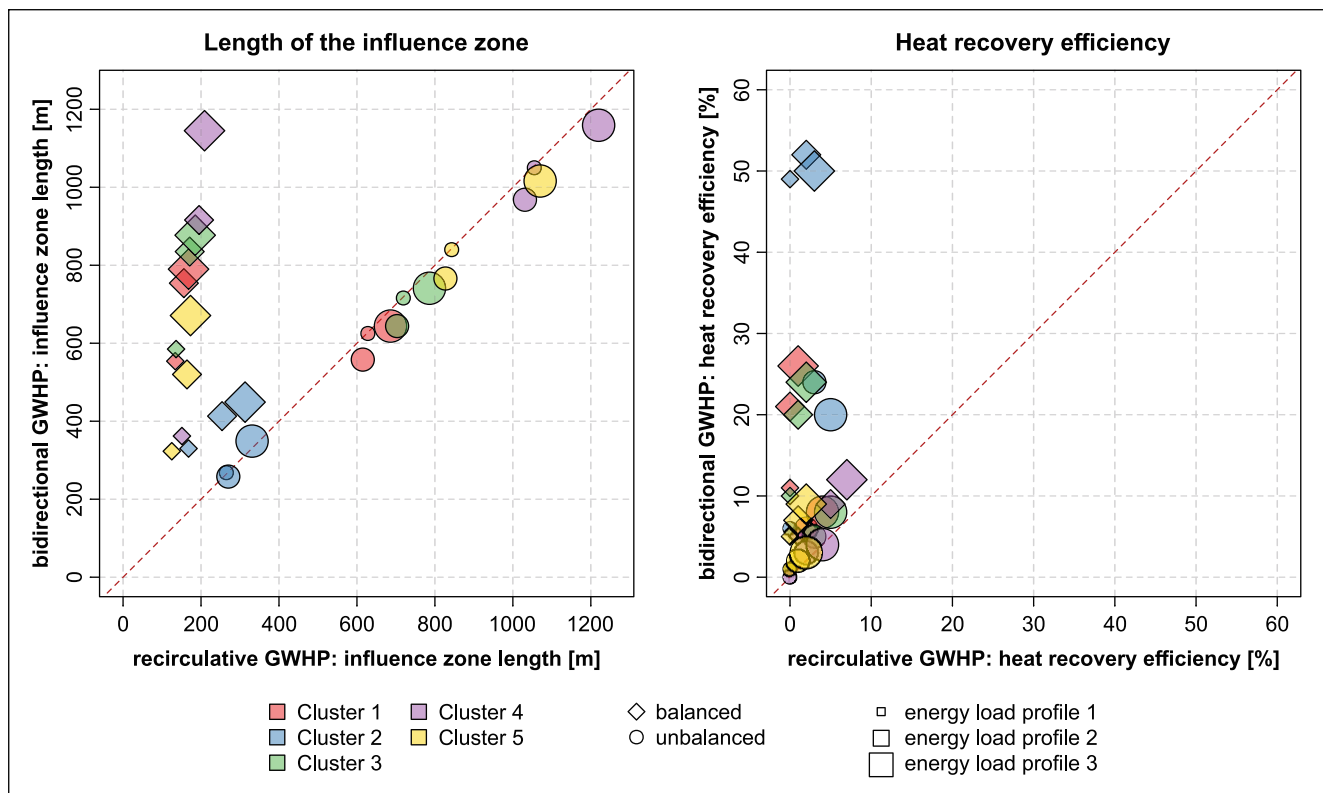
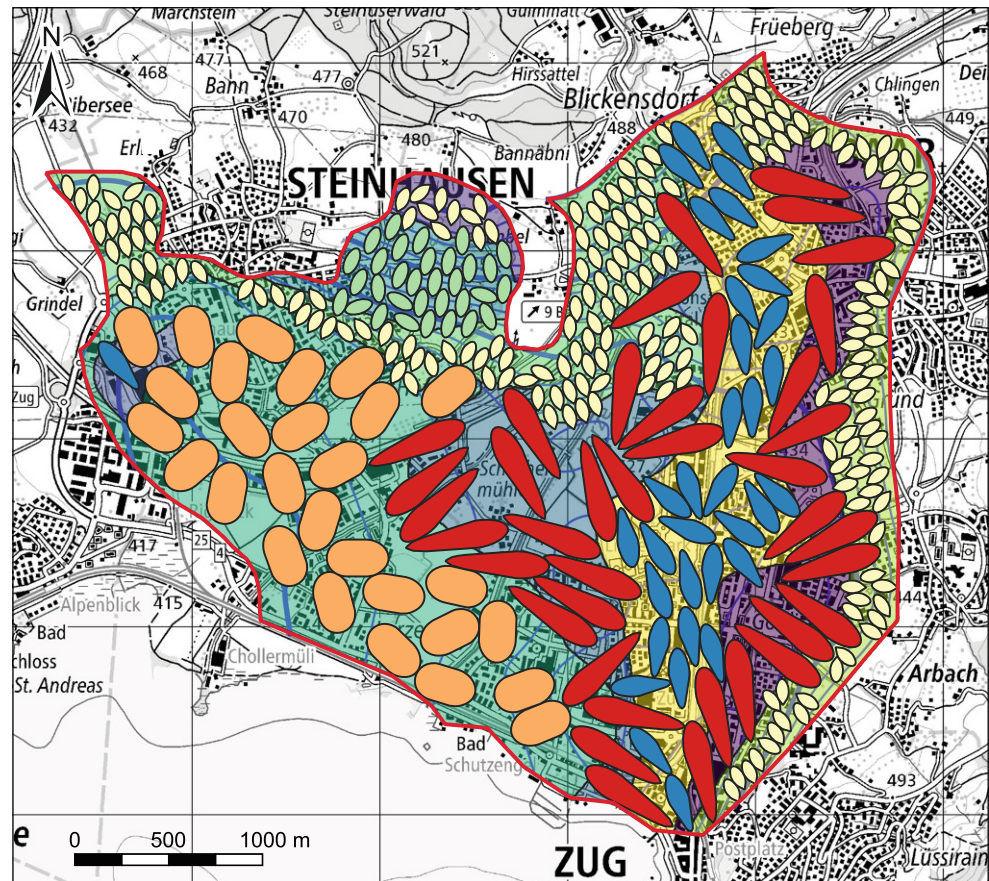


Fig. 4 a Length of the influence zones and (b) heat recovery efficiency for each cluster, as a function of the energy load profiles—whether balanced or unbalanced—and the operating modes (bidirectional or recirculative)

Abb. 4 a Länge der Einflusszonen und (b) Wärmerückgewinnungseffizienz für jeden Cluster in Abhängigkeit von den Energiebedarfsprofilen – ausgeglichen oder unausgeglichen – sowie den Betriebsmodi (bidirektional oder rezirkulierend)

Fig. 5 Spatial distribution of the influence zones for the balanced energy load profile 1

Abb. 5 Räumliche Verteilung der Einflusszonen für das ausgeglichene Energielastprofil 1



An example of the output of the packing algorithm used to quantify the heat use potential is shown in Fig. 5. In areas where the groundwater flow direction varies significantly throughout the domain, fewer GWHP doublets can be placed as their influence zones become larger.

GWHP heat use potential

The heat use potential and the specific heat use potential of the investigation perimeter for the three balanced and unbalanced energy load profiles are summarized in Table 3. These values fall approximately in the middle of the uncertainty ranges, which are significantly larger for the balanced energy load profiles because their influence zones are smaller. The heat use potential tends to be 2.5–3.5 times larger for GWHP doublets with a higher capacity (e.g., energy load profile 3) as well as 2–5 times larger for GWHP doublets with balanced profiles. Larger systems, with more extensive thermal influence zones, are more space-efficient because each well affects a greater aquifer volume, reducing the number of wells and land area required to meet the same heating and cooling demand.

Because of the mixture of GWHP doublet operating modes, an average heat recovery efficiency of approxi-

mately 10% can be achieved across the investigation area for medium and larger energy load profiles (see Fig. 4b).

Discussion

The presented methodology aiming at quantifying the GWHP heat use potential and its uncertainty has been applied to the deep aquifer, primarily focusing on open groundwater uses (GWHP doublets) and seasonal aquifer thermal energy storage (ATES).

According to the Canton of Zug, the cumulative maximum capacity of existing uses was previously estimated to be 4.3 MW for heat generation and 6.4 MW for cold generation (as of May 17, 2024). The average maximum capacity of existing GWHP doublets is approximately 240 kW, which roughly corresponds to energy load profile 2 (commercial and administrative building with a heating capacity of 250 kW). Assuming that the energy load profiles of existing GWHP doublets are unbalanced, an additional capacity of around 10 MW could still be installed.

The thermal footprint and the heat recovery efficiency of a GWHP doublet are directly influenced by the hydrogeological characteristics as well as by the balanced or unbalanced energy load profile. This highlights the challenge

Table 3 Uncertainty of the heat use potential in the study area for the three balanced and unbalanced energy load profiles. The value in parentheses corresponds to the originally determined heat use potential per cluster.

Tab. 3 Unsicherheit des Wärmenutzungspotenzials des Untersuchungsgebiets für die drei ausgeglichenen und unausgeglichenen Energielastprofile. Der Wert in der Klammer entspricht dem ursprünglich ermittelten Wärmenutzungspotenzial pro Cluster.

	Energy load profile 1	Energy load profile 2	Energy load profile 3
<i>Heat use potential (MW)</i>			
Balanced	16.08–23.22 (20.34)	23.5–40 (32.75)	33.75–56.25 (46.87)
Unbalanced	2–6 (4.2)	12.75–24 (16.75)	10.87–25.13 (15)
<i>Specific heat use potential (W/m²)</i>			
Balanced	1.42–2.05 (1.79)	2.07–3.52 (2.89)	2.98–4.95 (4.13)
Unbalanced	0.26–0.53 (0.37)	1.12–2.11 (1.48)	0.96–2.22 (1.32)

to determine one single technical heat use potential for an area. In fact, there is a range of possible technical heat use potentials that must be quantified and communicated to stakeholders, as they may impact future strategies. For example, the current study revealed that the potential is significantly greater for larger-capacity systems (e.g., 375 kW systems can yield 2.5 to 3.5 times higher potential than 60 kW systems) and for systems operating with balanced energy load profiles (which can achieve 3 to 5 times higher potential than unbalanced profiles). This finding implies that future GWHP development should prioritize larger-scale, integrated district heating/cooling solutions with seasonal balancing capabilities to maximize resource use and achieve greater energy impact. This has profound implications for urban planning, energy policy, and investment strategies, pushing towards district energy solutions that integrate heating and cooling demands over annual cycles.

A key strength of the approach lies in its practicality and computational efficiency, as it avoids simultaneous modeling of all GWHP doublets by classifying the aquifer groundwater into representative clusters. However, this simplification does not fully capture the complex heterogeneity of real-world aquifers, potentially leading to discrepancies between modeled and actual conditions (especially in the vicinity of the cluster boundaries). This is important because subsurface heterogeneity can have a significant impact on the thermal footprint (Di Dato et al. 2022). Furthermore, the influence zone (i.e., thermal plume) may be distorted by non-uniform groundwater flow patterns that are relatively marked in this study. To tackle these two limitations, more complex models in terms of heterogeneity and boundary conditions are needed.

The approach proposed to quantify the uncertainty on the heat use potential suffers the same limitations as mentioned above.

In this study, thermal dispersivity and its impact on the thermal footprint of GWHP doublets were not taken into

account, although thermal dispersivity has a strong impact (Park et al. 2015). Because dispersion is linearly proportional to velocity and dispersivity (Diersch 2014), more attention has to be given to dispersivity where groundwater flow velocities are significant (e.g., Park and Lee 2021). For small dispersivity values, the thermal plume is more compact and therefore the heat recovery efficiency is more sensitive to changes in hydraulic gradient direction.

Another key uncertainty is the energy load profile itself. Operational adjustments can play a crucial role in mitigating thermal interference and optimizing heat recovery. This includes dynamically adjusting groundwater flow rates and/or the temperature changes of the injected water to meet thermal energy demands while minimizing thermal impacts. This requires a monitoring of the groundwater temperature and most probably a regulatory framework.

The numerical mesh of the box models affects the size of the influence zone and, consequently, the GWHP heat-use potential. The number of layers and elements can be increased until the simulation results converge. However, the required high resolution renders the proposed workflow impractical due to the increased computing time. Considering the uncertainties mentioned above, the numerical uncertainty may be negligible.

Future improvements could focus on several aspects. First, conducting more comprehensive sensitivity analyses beyond the currently investigated parameters would provide a deeper understanding of the relationships between hydrogeological conditions and the size of influence zones. We assume that thermal dispersivity may play a critical role, particularly because it is difficult to reliably estimate from field data and cannot be determined through simple field experience (like pumping tests) alone. Second, investigating the “trade-off” between the size of the influence zone and the efficiency of heat recovery in relation to the system type and energy load profile is crucial for optimizing system design and placement. Third, accounting for non-

uniform groundwater flow could be efficiently achieved by upgrading the packing algorithm with the mapping of the influence zone on the non-uniform flow when the influence zones are positioned. Finally, the methodology, based on a limited number of complex numerical simulations, can potentially be simplified and accelerated by a comparison with analytical solutions (e.g., linear advective heat transport model; Pophillat et al. 2020; Halilovic et al. 2023). This highlights a potential area for future improvement, as analytical solutions could offer quicker initial assessments or serve as benchmarks for the more complex numerical models.

Conclusion

This study provides a comprehensive assessment of the heat use potential of the deep aquifer in the Baar-Zug-Steinhausen area (Switzerland), employing a flexible and computationally efficient methodology. By classifying the aquifer into representative clusters and simulating various energy load profiles, the research determined the maximum technical heat use potential while ensuring adherence to regulatory thermal impact limits. The findings highlight that the deep groundwater resource is significantly underutilized, with the potential for installed capacity to be tripled, especially for larger systems and balanced energy profiles.

While the methodology offers a robust framework for such assessments, leveraging numerical simulations and a packing algorithm, it also includes simplifications, such as assuming homogeneous subsurface conditions and specific doublet configurations. The quantification of uncertainties, albeit a simplified approach, provides valuable insight into the range of potential outcomes, reinforcing the reliability of the determined use potential. Future research should aim to refine these uncertainties through more comprehensive sensitivity analyses, explore the trade-offs between influence zone size and heat recovery efficiency, and potentially integrate analytical solutions to further enhance the methodology's precision and applicability. This research serves as a crucial foundation for sustainable thermal energy planning and development in the region, paving the way for optimized and environmentally sound groundwater heat use.

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